

Scalar k:2k covariance-simplex compactification, uniform projective

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-28 · this version issued 2026-07-28 · v1.0

Purpose and scope

The unbounded mixed scalar $k:2k$ parameter space is reduced exactly to one compact covariance simplex and one explicit compact semialgebraic (x, τ) domain with a rigorous radial tail. The projective boundary is uniformly safe and its first two inverse-amplitude corrections are nonnegative, and strict for positive projective coordinate. The remaining strict mixed core still requires directed-rounding interval certification; the mixed all- q theorem and Sector A are not closed.

R-111 used $a = A^2/v$ and $r = w/v$. Those coordinates are exact when $v > 0$, but $r \rightarrow \infty$ obscures the second covariance face. This note replaces them by the closed simplex

$$c = \frac{v}{v + 4w}, \quad s = \frac{4w}{v + 4w}, \quad c + s = 1.$$

The complete packet, covariance square, projective limit, and radial tails then extend continuously to both faces.

An exact sum of squares gives a global packet floor, and three exact positive identities give a uniform covariance-square floor. Therefore the exterior of

$$(2x + \tau)^2 \leq 32x + 20\tau, \quad 0 \leq c \leq 1, \quad 0 \leq x \leq 8, \quad 0 \leq \tau \leq 20$$

is already proved. On this compact domain the complete radial tail outside $T, U \leq 50$ is below 1.1×10^{-17} .

On the projective boundary the gap has a uniform expansion through order b^{-2} with remainder $O(x^3 b^{-3})$. The leading gap, the b^{-1} coefficient, and the b^{-2} coefficient are nonnegative; the second one has an exact degree-twelve Bernstein certificate and retains one quarter of the strict R-110 local margin. The third coefficient is negative at one exact rational point. Thus positivity of every projective coefficient is a false shortcut, not a counterexample to the target.

R-103 REG and R-087 fixed-cutoff variational CORE retain their accepted owners. Nothing here proves the full A1 one-fresh-root cluster, the separate one-use source/sextic aggregation, OVERLAP_{src}, Nelson, removals, an interacting measure, or Sector-A closure. Tier remains T4.

All polynomial and rational identities below are exact. The formal numerical scans used to choose adversarial fixtures are not evidence and are not cited. The only displayed decimal is an exact analytic tail expression evaluated at the declared truncation radius.

1 Closed covariance-simplex normal form

Let $\gamma = v + 4w > 0$ and set

$$c = \frac{v}{\gamma}, \quad s = \frac{4w}{\gamma}, \quad b = \frac{A^2}{\gamma}, \quad \tau = q\gamma^2, \quad x = b\tau. \quad (1.1)$$

Thus $c, s \geq 0$ and $c + s = 1$. Let T, U be independent unit exponentials and put

$$\rho = \frac{cT}{2}, \quad \sigma = \frac{sU}{8}. \quad (1.2)$$

At $c = 0$ or $s = 0$ the zero-weight exponential is a harmless dummy variable. The relative phase ϕ is uniform. With

$$L = \rho + 4\sigma - \frac{1}{2}, \quad Q = \rho^2 + 10\rho\sigma + 4\sigma^2 - \rho - \sigma, \quad B = 6\rho\sqrt{\sigma} \cos \phi, \quad (1.3)$$

the exact R-110 packet and covariance square become

$$F := \frac{P}{\gamma^2} = bL + \sqrt{b}B + Q, \quad (1.4)$$

$$K := \frac{H}{\gamma^4} = b^2K_2 + bK_1 + K_0, \quad (1.5)$$

where

$$K_2 = \frac{c^2 + s^2}{2}, \quad (1.6)$$

$$K_1 = c^3 + \frac{5}{2}c^2s + cs^2 + \frac{s^3}{4}, \quad (1.7)$$

$$K_0 = \frac{5}{4}c^4 + \frac{1}{4}c^3s + \frac{25}{16}c^2s^2 + \frac{1}{4}cs^3 + \frac{5}{64}s^4. \quad (1.8)$$

The packet is centered. The exact local margin is

$$\Delta(c, s) := K - 2 \text{Var}(F) = \frac{c^4}{4} + \frac{c^2s^2}{4} + \frac{s^4}{64}. \quad (1.9)$$

It has the elementary uniform lower bound

$$64\Delta = (4c^2 + s^2)^2 + 8c^2s^2, \quad 4c^2 + s^2 = \frac{4}{5} + 5\left(c - \frac{1}{5}\right)^2, \quad \Delta \geq \frac{1}{100}. \quad (1.10)$$

The phase-integrated Laplace transform is best written directly in the compact variables:

$$M(c, x, \tau) = \mathbb{E}\left[e^{-xL - \tau Q} I_0(6\sqrt{x\tau} \rho\sqrt{\sigma})\right]. \quad (1.11)$$

The desired scalar theorem is

$$\log M(c, x, \tau) \leq \frac{x^2K_2 + x\tau K_1 + \tau^2K_0}{4}. \quad (1.12)$$

For finite b one has $x = b\tau$. Allowing $\tau = 0$ and $x > 0$ adds the large-amplitude projective boundary and makes the parameter space compact.

2 Exact global exterior and a compact interval core

Minimizing the phase in (1.4) gives

$$F_- = \rho^2 + 10\rho\sigma + 4\sigma^2 + (b-1)\rho + (4b-1)\sigma - \frac{b}{2} - 6\sqrt{b}\rho\sqrt{\sigma}. \quad (2.1)$$

Theorem 2.1 (global floor and compact semialgebraic reduction)

For every $b, \rho, \sigma \geq 0$,

$$\boxed{F_- = \left(\rho - \frac{1}{2}\right)^2 + 4\left(\sigma - \frac{1}{8}\right)^2 + \rho\left[(\sqrt{b} - 3\sqrt{\sigma})^2 + \sigma\right] + 4b\sigma - \frac{b}{2} - \frac{5}{16}.} \quad (2.2)$$

Hence

$$F \geq -\frac{b}{2} - \frac{5}{16}. \quad (2.3)$$

On the covariance side,

$$K_2 = \frac{1 + (c-s)^2}{4}, \quad (2.4)$$

$$K_1 = \frac{1}{4} + \frac{c(1+5c-3c^2)}{4}, \quad (2.5)$$

$$K_0 = \frac{1}{16} + \frac{s^4 + 76c^2s^2 + 76c^4}{64}. \quad (2.6)$$

Every remainder is nonnegative on $c + s = 1$, so

$$\boxed{K \geq \frac{(2b+1)^2}{16}.} \quad (2.7)$$

The floor proves (1.12) whenever

$$\tau \geq R(b) := \frac{32b+20}{(2b+1)^2}. \quad (2.8)$$

Indeed $\log M \leq \tau(b/2 + 5/16)$, and the right side of (1.12) is at least $\tau^2(2b+1)^2/64$.

Moreover,

$$20(2b+1)^2 - (32b+20) = 16b(5b+3) \geq 0, \quad (2.9)$$

$$8(2b+1)^2 - b(32b+20) = 4(3b+2) > 0. \quad (2.10)$$

Thus only $0 \leq \tau \leq 20$ and $0 \leq x = b\tau \leq 8$ remain. Eliminating $b = x/\tau$ gives the exact closed residual condition

$$\boxed{(2x + \tau)^2 \leq 32x + 20\tau.} \quad (2.11)$$

This proves the claimed compactification. $\square$

Theorem 2.2 (uniform complete radial tail)

Using $I_0(z) \leq e^z$ and the same squares,

$$\begin{aligned} xL + \tau Q - 6\sqrt{x\tau}\rho\sqrt{\sigma} \\ = \tau(\rho^2 - \rho) + \tau(4\sigma^2 - \sigma) - \frac{x}{2} + 4x\sigma \\ + \rho\left[(\sqrt{x} - 3\sqrt{\tau\sigma})^2 + \tau\sigma\right] \geq -\frac{x}{2} - \frac{5\tau}{16}. \end{aligned} \quad (2.12)$$

Therefore the contribution of $\{T > R\} \cup \{U > R\}$ to the complete phase/Bessel integral obeys

$$\boxed{M_{\text{tail}}(R) \leq e^{x/2+5\tau/16} \left(2e^{-R} - e^{-2R}\right).} \quad (2.13)$$

On (2.11), $R = 50$ gives

$$M_{\text{tail}}(50) < 1.091 \times 10^{-17}. \quad (2.14)$$

This is an analytic tail bound, not quadrature convergence evidence. $\square$

3 Uniform projective boundary

For $b > 0$, let $\delta = b^{-1} = \tau/x$ and keep x fixed. Then

$$\tau F = x(L + \sqrt{\delta} B + \delta Q). \quad (3.1)$$

Under the leading tilt, T and U remain independent exponentials with rates

$$\alpha = 1 + \frac{cx}{2}, \quad \beta = 1 + \frac{sx}{2}. \quad (3.2)$$

The limiting transform and gap are

$$M_{\infty} = \frac{e^{x/2}}{(1 + cx/2)(1 + sx/2)}, \quad (3.3)$$

$$\boxed{D_0 = d(cx/2) + d(sx/2), \quad d(u) = \frac{u^2}{2} - u + \log(1 + u).} \quad (3.4)$$

Since $d'(u) = u^2/(1 + u)$, $D_0 > 0$ for $x > 0$. On $0 \leq x \leq 8$,

$$d(u) \geq \frac{u^3}{15} \quad (0 \leq u \leq 4), \quad c^3 + s^3 \geq \frac{1}{4},$$

and hence

$$\boxed{D_0 \geq \frac{x^3}{480}.} \quad (3.5)$$

Theorem 3.1 (uniform expansion through the second correction)

Uniformly for $c + s = 1$ and $0 \leq x \leq 8$,

$$\boxed{G_b := \frac{\tau^2 K}{4} - \log M = D_0 + \frac{D_1}{b} + \frac{D_2}{b^2} + O\left(\frac{x^3}{b^3}\right).} \quad (3.6)$$

The first coefficient is

$$\boxed{D_1 = \frac{x^3 N_1(c, s, x)}{16(2 + cx)^2(2 + sx)^2} \geq 0,} \quad (3.7)$$

where

$$\begin{aligned} N_1 = & 4c^5 s^2 x^3 + 16c^5 s x^2 + 16c^5 x + 10c^4 s^3 x^3 + 56c^4 s^2 x^2 + 104c^4 s x + 64c^4 \\ & + 4c^3 s^4 x^3 + 56c^3 s^3 x^2 + 172c^3 s^2 x + 152c^3 s \\ & + c^2 s^5 x^3 + 20c^2 s^4 x^2 + 88c^2 s^3 x + 112c^2 s^2 \\ & + 4cs^5 x^2 + 32cs^4 x + 32cs^3 + 4s^5 x + 16s^4. \end{aligned} \quad (3.8)$$

All coefficients are positive.

The second coefficient satisfies the stronger exact bound

$$\boxed{D_2 \geq \frac{x^2 \Delta(c, s)}{4} \geq \frac{x^2}{400}.} \quad (3.9)$$

Uniform remainder. Write $\varepsilon = b^{-1/2}$. For $0 < \theta \leq 1$ and $b \geq 10$, put $b_\theta = (\theta\varepsilon)^{-2} \geq 10$. The intermediate exponent $L + \theta\varepsilon B + \theta^2\varepsilon^2 Q$ is F_{b_θ}/b_θ . The sharper phase-minimum identity

$$b_\theta - 1 + 10\sigma - 6\sqrt{b_\theta\sigma} = 10 \left(\sqrt{\sigma} - \frac{3\sqrt{b_\theta}}{10} \right)^2 + \frac{b_\theta}{10} - 1 \geq 0$$

gives $F_{b_\theta} \geq -b_\theta/2$ and hence the common dominator $e^{x/2}e^{-T-U}$; the endpoint $\theta = 0$ follows by continuity. The sixth ε derivative is a Bell polynomial in $-xB - 2x\varepsilon Q$ and $-2xQ$. Every monomial has x -degree at least three, and B, Q have simplex-uniform polynomial coefficients. Thus the integrated sixth derivative is bounded by

$$Cx^3(1+T+U)^N e^{-T-U} \quad (3.10)$$

for one finite constant on $x \leq 8$. Phase symmetry removes odd powers.

If

$$A_1 = \mathbb{E}_x \left[\frac{x^2 B^2}{2} - xQ \right], \quad (3.11)$$

$$A_2 = \mathbb{E}_x \left[\frac{x^4 B^4}{24} - \frac{x^3 B^2 Q}{2} + \frac{x^2 Q^2}{2} \right], \quad (3.12)$$

then

$$D_1 = \frac{x^2 K_1}{4} - A_1, \quad D_2 = \frac{x^2 K_0}{4} - \left(A_2 - \frac{A_1^2}{2} \right). \quad (3.13)$$

The centered identity $\mathbb{E}_0 Q = 0$ and smooth tilted moments give $A_1 = O(x^2)$ and $A_2 = O(x^2)$. Thus the logarithmic sixth-order combinations $A_1 A_2$ and A_1^3 are respectively $O(x^4)$ and $O(x^6)$, hence $O_8(x^3)$. Together with (3.10), this proves that passing from the raw transform to its logarithm preserves the factored $O(x^3 b^{-3})$ remainder.

Exact exponential moments give (3.7). After $s = 1 - c$, the exact positive denominators for D_2/x^2 and $(D_2 - x^2 \Delta/4)/x^3$ are respectively

$$256(2+cx)^4(2+(1-c)x)^4, \quad 64(2+cx)^4(2+(1-c)x)^4.$$

They are strictly positive on $0 \leq c \leq 1, x \geq 0$. Clear these denominators and express both numerators in the degree-twelve Bernstein basis in c . Each of the thirteen Bernstein coefficients is coefficientwise nonnegative in x . The minimum power coefficients for the margin reserve are

$$16, \frac{8}{3}, \frac{4}{11}, \frac{2}{55}, \frac{1}{495}, \frac{1}{198}, \frac{1}{44}, \frac{1}{198}, \frac{16}{495}, \frac{32}{55}, \frac{64}{11}, \frac{128}{3}, 256. \quad (3.14)$$

This is a finite exact polynomial certificate. Equations (3.10)–(3.14) prove (3.6)–(3.9). $\square$

Corollary 3.2 (existential uniform large-amplitude theorem)

There exists a finite universal b_* such that, for every covariance-simplex point and every $\tau \geq 0$, (1.12) holds whenever $b \geq b_*$. On $x \leq 8$, (3.5)–(3.9) and the factored remainder prove the result for all sufficiently large b , uniformly down to $x = 0$. For $x \geq 8$, the sharper consequence of (2.1),

$$b \geq 10 \implies F \geq -\frac{b}{2}, \quad (3.15)$$

and $K_2 \geq 1/4$ close the target directly. The theorem is analytic but non-effective because the constant in (3.10) has not yet been extracted. $\square$

4 Origin and face patches

Theorem 4.1 (rigorous origin cusp)

Let $X_0 = -F$. From (2.3),

$$X_0 \leq B_0 = \frac{b}{2} + \frac{5}{16}, \quad \text{Var}(F) = \frac{K - \Delta}{2}. \quad (4.1)$$

The one-sided Bernstein MGF inequality gives

$$\log \mathbb{E} e^{\tau X_0} \leq \frac{\tau^2(K - \Delta)}{4(1 - \tau B_0/3)}. \quad (4.2)$$

Consequently (1.12) holds whenever

$$\tau B_0 \leq \frac{3\Delta}{K}. \quad (4.3)$$

In compact coordinates this is the exact polynomial condition

$$(8x + 5\tau)(x^2 K_2 + x\tau K_1 + \tau^2 K_0) \leq 48\Delta\tau^2. \quad (4.4)$$

This supplies a certified cusp at the equality point $(x, \tau) = (0, 0)$. $\square$

Proposition 4.2 (face patch and strict residual interior)

R-111 proves (1.12) strictly for $\tau > 0$ on $c = 0$ and $c = 1$. Remove the open interiors of the origin set certified by Theorem 4.1 and the projective set certified by Corollary 3.2. Any closed residual slice separated from their common equality boundaries is compact after the tail (2.13) is removed, and the complete integral (1.11) is continuous there. Strictness on each face and a finite-subcover argument then supply positive face neighborhoods on that slice. Equivalently, after a finite closed-slice decomposition, the unresolved interval problem is confined to strict compact interiors in c, x, τ . No single uniform face width across the equality origin is claimed. The slice widths are presently existential rather than machine-certified, so this proposition is a reduction, not the mixed all- q theorem. $\square$

5 A necessary failed route at third order

Proposition 5.1 (all-order coefficient positivity is false)

The order b^{-3} coefficient is

$$D_3 = - \left(A_3 - A_1 A_2 + \frac{A_1^3}{3} \right), \quad (5.1)$$

where

$$A_3 = \mathbb{E}_x \left[-\frac{x^3 Q^3}{6} + \frac{x^4 B^2 Q^2}{4} - \frac{x^5 B^4 Q}{24} + \frac{x^6 B^6}{720} \right]. \quad (5.2)$$

At the exact rational simplex point

$$c = \frac{3}{5}, \quad s = \frac{2}{5}, \quad x = \frac{24}{25}, \quad (5.3)$$

one obtains

$$D_3 = -\frac{627811338105359170693920}{190578044621571595050427561} < 0. \quad (5.4)$$

At the same point D_0, D_1, D_2 are positive. Thus (5.4) retires only the claim that every inverse-amplitude coefficient is nonnegative. It is not a counterexample to (1.12). This is registered as

$$\text{NG-2026-07-28-A13-K2K-ALL-ORDER-PROJECTIVE-COEFFICIENT-POSITIVITY}. \quad (5.5)$$

6 Directed-rounding successor

After Theorems 2.1–4.1 and the exact R-111 faces, the remaining scalar task is finite. A suitable certificate must:

1. use the compact coordinates (c, x, τ) and the constraint (2.11);
2. keep the combined phase exponent rather than bounding a huge isolated I_0 value on the radial box;
3. use (2.13) at $R = 40$ or 50 for the exterior;
4. use Theorem 4.1 near the origin and Theorem 3.1 near $\tau = 0$;
5. use the exact R-111 faces near $c = 0, 1$;
6. apply directed-rounding Taylor or interval cubature only to the strict residual interior.

For fixed U and phase θ , the T integral has the stable exact form

$$\int_0^\infty e^{-aT^2-dT} dT = \frac{\sqrt{\pi}}{2\sqrt{a}} \operatorname{erfcx}\left(\frac{d}{2\sqrt{a}}\right), \quad (6.1)$$

with its continuous $a \downarrow 0$ limit when $d > 0$; that condition holds on the intended $\tau = 0$ and $c = 0$ degeneracies after the linear terms are combined. This reduces validated cubature to (U, θ) and avoids the separate exponential growth of I_0 . Arb or an equivalent directed-rounding backend is the recommended next implementation. No finite Gauss–Laguerre scan is promoted to evidence here.

Even a successful certificate for (1.12) would close only the stationary scalar normalizer. The next production theorem would still be the complete one-fresh-root, contraction-connected A1 cluster retaining heat, mean, covariance trace, mixed baseline, rational recovery, every visit, random- W double divergence, and the R-063 forest once. The source energy and terminal sextic must then be aggregated in a separate one-use ledger.

7 Devil’s-advocate review

1. **Objection: the $r \rightarrow \infty$ face destroys the projective remainder.**

DISMISSED. The simplex coefficients are continuous at $c = 0$ and $s = 0$. Dummy unit exponentials carry zero weight, and the exact common floor supplies the derivative dominator on the closed simplex.

2. **Objection: $D_0 = O(x^3)$ can be swamped at $x \rightarrow 0$.**

VALID WITH MITIGATION. A bare $D_0 + D_1/b + O(b^{-2})$ statement is insufficient. The proof retains D_2 , certifies $D_2 \geq x^2\Delta/4$, and factors the remainder as $O(x^3b^{-3})$. The finite- b origin uses the separate Bernstein-MGF cusp.

3. **Objection: the projective expansion is being exchanged with $x \rightarrow \infty$.**

UPHELD AGAINST THAT EXCHANGE. The expansion is used only on $x \leq 8$. The exact floor closes the exterior before the expansion is invoked.

4. **Objection: every projective coefficient is positive.**

UPHELD AS FALSE. Equation (5.4) is an exact negative third coefficient. The proof uses a remainder enclosure, not sign induction.

5. **Objection: the compact box itself proves the theorem.**

UPHELD AS AN OVERCLAIM. Compactness and a small tail make a finite certificate possible; they do not supply it. The strict mixed interior is still open.

6. **Objection: signs, factors, units, or hard-coded decimals hide the result.**

DISMISSED FOR THE CLAIMED SCOPE. Two non-importing scripts derive the quotient, K_2, K_1, K_0 , both Bernstein tables, the floor identities, and the rational D_3 fixture. The tail decimal is recomputed from its exact formula and declared radius. The exploratory scans are excluded from the evidence contract. External adversarial review of the interval successor is explicitly invited.

8 Authorities and scope firewall

This note depends directly on R-110 for the exact stationary packet and on R-111 for the degenerate-face theorem and the first fixed-chart projective calculation. R-063, R-087, R-101, R-103–R-105, and R-108–R-111 retain the production owner map.

Proved here are the exact simplex quotient, Theorems 2.1–2.2, Theorem 3.1, Corollary 3.2, Theorem 4.1, Proposition 4.2, and the method no-go in Proposition 5.1. Not proved are a directed-rounding certificate for the strict mixed core, the genuinely mixed scalar all- q theorem, an effective b_* , a full A1 embedding, the owner-preserving production cluster, the one-use source/sextic aggregation, $\text{OVERLAP}_{\text{src}}$, Nelson, removals, an interacting measure, Sector-A closure, or a tier promotion.

9 Executable verification

The primary certificate reconstructs the compact packet from radial variables, transforms H and the local margin, computes the first three projective coefficients, and proves both degree-twelve Bernstein tables by exact rational arithmetic. The independent certificate imports no primary code: it expands

$$e^{-x\delta Q} I_0(x\sqrt{\delta} B)$$

coefficient by coefficient and reconstructs the logarithm separately. It also independently checks the global sum of squares, compact domain, tail, origin polynomial, and rational third-order failure.

The primary contract is 69/69 and the non-importing independent contract is 50/50. The manifest-pinned verifier reruns both children and checks source, result, authority, PDF, ledger, negative, exploration, and scope contracts.

Result footer

Result ID: A13-CLASSII-SCALAR-K2K-COVARIANCE-SIMPLEX-UNIFORM-PROJECTIVE-COMPACT-CORE-REDUCTION

Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

Ledger ID: R-112

Status: T4 ANALYTIC/EXACT/EXECUTED ADVANCE AND METHOD BOUNDARY;
MIXED ALL-Q, NELSON, AND SECTOR A OPEN

Scope: Exact stationary scalar circle k:2k model inherited from R-110.
Closed covariance simplex and compact (c,x,tau) reduction; no
full A1 production embedding.

Precise statement: the unbounded mixed scalar parameter space has an exact
compact semialgebraic reduction with a rigorous radial tail; the
projective boundary has a uniform factored expansion, positive
 D_0, D_1, D_2 , and an existential all- q large-amplitude theorem; D_3 can
be negative, so all-order coefficient positivity is false.

Dependencies: R-063, R-087, R-101, R-103--R-105, R-108--R-111.

Evidence grade: ANALYTIC/EXACT/EXECUTED; TWO NON-IMPORTING ROUTES.

Reproduction command:

```
E:\Dev\TECT.venv\Scripts\python.exe '
codes/foundations/a13_classii_scalar_k2k_covariance_simplex_uniform_compact_core_verify.py
```

Expected output:

primary 69/69; independent 50/50; manifest-pinned integrated and
aggregate contracts; PASS.

Negative record:

NG-2026-07-28-A13-K2K-ALL-ORDER-PROJECTIVE-COEFFICIENT-POSITIVITY.

Falsification gate: failure of any simplex, scaling, local-margin, projective,
Bernstein, remainder, floor, compact-domain, radial-tail, origin,
third-coefficient, independent, hash, PDF, ledger, or scope check.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement:

no strict mixed-core interval certificate, mixed all-q theorem,
full A1 cluster, OVERLAP_src, Nelson, removals, measure, or Sector
A closure is proved.

Next required action: implement a directed-rounding interval certificate on
the strict compact core using the analytic origin, face,
projective, high-q, and radial-tail patches; only after scalar
closure attempt the owner-preserving full-A1 conditional cluster
and separate one-use source/sextic aggregation ledger.